ES 673: NeuroAI

Exploring Non-Backpropagation Learning Paradigms

- **Due Date:** September 11th, 2026, 10 PM
- **Deliverables:** Code Implementation (Python/PyTorch), Toy Dataset Experiments, Methodological Explorations, and a Short Report. Kindly submit your code files, report (as pdf) on Classroom.
- **AI Usage:** Generative AI models could be used for code implementation but kindly make sure you understand all the aspects of the code. The report should be written yourself. Marks would be deducted if found otherwise.

Backpropagation is the cornerstone of modern deep learning, yet it faces well-documented biological implausibilities (the weight transport problem, the locking of forward/backward passes, and lack of local layer-wise autonomy). In this assignment, you will go through two novel non-standard learning paradigms:

1. **The Forward-Forward Algorithm (FF)**: Replaces standard forward/backward phases with dual forward passes (*positive* and *negative*) optimizing a local "goodness" function layer-by-layer.

2. **NoProp**: Eliminates both full forward and full backward propagation by framing layer training independently through local denoising, drawing inspiration from diffusion and flow-matching mechanics.

Part A: The Forward-Forward Algorithm (Hinton, 2022)

Reference Paper: The Forward-Forward Algorithm: Some Preliminary Investigations

Core Implementation

- **Dataset:** Use a toy dataset like MNIST or a synthetic 2D multi-class dataset.
- **Positive/Negative Data Creation:** Implement the data-masking or negative-data generation procedure described in the paper.
- **Goodness Function & Layer Training:** Implement the basic functionality of the model using the goodness function given in the paper.

Exploration & Modification

- **Alternative Goodness Metrics:** Test an alternative goodness function (e.g., incorporating negative sums of squared activities, sparse activation penalties, or cosine similarity between layer activations) and analyze how it impacts feature separation.

Part B: NoProp Learning (Li et al., 2025)

Reference Paper: NoProp: Training Neural Networks without Full Back-propagation or Full Forward-propagation

Core Implementation

- **Architecture Setup:** Construct a multi-layer network where blocks/layers do not rely on standard chained forward activations propagating sequentially from input to output during training.
- **Local Denoising / Target Matching:** Implement the independent layer-wise training objective inspired by diffusion or flow-matching, where local layers learn to map noisy inputs to target representations independently.

Part C: Comparative Analysis & Discussion

In your final report, provide an empirical and conceptual comparison addressing the following questions:

1. **Computational Efficiency & Scaling:** How do training times, memory footprints (avoiding activation storage for backprop), and hardware distribution potentials compare between backprop, Forward-Forward, and NoProp on your toy datasets?

2. **Representational Differences:** Backprop naturally builds a strict hierarchy of abstraction. How do the representations learned layer-by-layer in FF and NoProp differ?

3. **Biological Plausibility:** Discuss how closely these algorithms resolve the biological constraints of real cortical networks compared to standard backpropagation.

Deliverables & Submission Guidelines

1. **Codebase:** A clean, well-commented GitHub repository (or Jupyter Notebooks) containing your implementations, training loops, and modification experiments.
2. **Short Report** highlighting your implementation and findings with the following tentative flow:
    - Introduction to local learning vs. global backprop.
    - Implementation details of your FF and NoProp pipelines.
    - Results, visualizations
    - Critical discussion on the future of backprop-free architectures in NeuroAI.